

Information technologies and human rights

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May 2024

To be published in *Technology and Equality. Shifting Roles in a Complex Relationship*.
Editors Sven Ove Hansson and Colleen Murphy. Rowman and Littlefield.

Abstract

In this chapter, a broad framework is proposed for assessing the significance of ITs for the equal protection of human rights. We will first examine the moral and legal dimension of human rights in connection with technological developments and their impacts on equality. We will then consider the way in which IT affects the fundamental human rights principles—dignity, liberty, and equality—and finally the way in which IT challenges some specific human rights. On this basis, the argument is made that human rights do not have a merely protective role to play in the face of the novel threats posed by IT: they also have an aspirational role, offering a blueprint in light of which to set out the kinds of goals we would want to achieve in shaping a good IT-based society. While they are incomplete as such a blueprint, they can offer good guidance.

1. Introduction

In the span of just a few decades we have witnessed three major social transformations driven by information technologies (ITs): first came the computer revolution, bringing information machines to all domains of human life; then came the Internet revolution, connecting such machines all over the world through interlinked high-speed communication networks; and finally the AI revolution has delivered automata capable of engaging in a vast and rapidly increasing range of cognitive activities. The combination of huge computational power, vast networked digital data resources, and intelligent technologies has led to an ever-accelerating stream of innovations: search engines; social networks; robotic factories; intelligent digital assistants; self-driving cars and airplanes; and systems that speak, understand, and translate language; make predictions; take autonomous decisions, generate texts, images, films and multimedia content.

Most social functions are now carried out through socio-technical systems—i.e., social arrangements combining humans and technologies—where ITs play a large and growing role, shaping all domains of social and individual life: from the economy to politics and administration, from work to socialisation and leisure. This has started in Western countries but has rapidly expanded to cover the entire world, as ITs themselves facilitate global access to science and technology. More than 50 billion interacting digital devices now exist, distributed across the earth and increasingly also in space. They process an

amount information many times larger than all the information produced by humanity before entering the digital era.

The present and prospective impacts of ITs on individuals and society raise appreciations and even enthusiasms, as well as worries and even fears. It is often observed that ITs, like many other technologies, may be looked at from a dual perspective, focusing on the positives as well as on the negatives. On the one hand, they enhance human abilities; improve security and efficiency, facilitate communication and access to information; support research; and contribute to better education, health care, and environmental protection. On the other hand, they may enable control, manipulation, and discrimination; disrupt social interactions; and expose humans to harm resulting from technological failures or disregard for individual rights and social values.

In the following we shall examine the significance of human rights for the equitable development and deployment of ITs. First, in Section 2, we will discuss the intersection between human rights and technological challenges. Then, in Section 3, we will discuss the ways in which ITs are impacting the fundamental human rights principles, namely, dignity, liberty, and equality. Finally, in Section 4, we will turn to some specific human rights.

2. Human rights

In this section, we shall consider the moral and legal dimensions of human rights and their significance as a framework for addressing the risks of ITs and promoting the equitable enjoyment of the opportunities they offer.

2.1. Human rights as moral rights

A broad conception of human rights is provided by Amartya Sen (2004), according to whom human rights are primarily moral demands concerning individual freedoms, i.e., opportunities that should be accessible to everybody. On this conception, human rights do not include only negative liberties, whose enjoyment mainly requires noninterference from third parties, as is the case with freedom of movement or of expression. They also include positive liberties, which require the provision of resources, as is the case with the social rights to education or health. On Sen's account, not all opportunities in which humans may be interested qualify as human rights, but only those opportunities which are both especially important for the individuals concerned and socially influenceable. In this connection, technologies are highly relevant, not only as threats but also as facilities that enable societies to expand existing meaningful opportunities and even create new ones, thus broadening the scope and dimensions of human rights.

Human rights as moral demands give rise to moral obligations for individuals, groups, and organisations, namely, the negative obligation not to interfere in others' enjoyment of such rights, and the positive obligation to contribute to the goal of enabling such enjoyment in the ways that are suitable for each agent, taking into account the constraints and commitments that apply to their situation. These obligations may have different degrees of stringency, for different addressees, depending on the different rights and social roles in

question: some are strict obligations, to be always complied with, and are never defeated or only defeated under exceptional circumstances (this applies in particular to the prohibitions on actively interfering with the enjoyment of rights); others just concern the need to take the right properly into account in deliberation, with an understanding of what it entails (this goes in particular for positive obligations to provide the means to exercise rights).

2.2. Human rights as legal rights

Though human rights as moral requirements go beyond legal prescriptions, human rights need to have legal recognition if they are to be implemented through legal systems. Following a Kantian argument (Kant 1996, 456), it may indeed be argued that there is a moral obligation to ensure that those aspects of moral human rights that can only, or preferably, be implemented through legal means are written into law. These are aspects that are to be realised through binding and enforceable legal obligations applying to public and private actors, or through direct or supportive action by state apparatuses.

The translation of moral human rights into legal requirements takes different parallel paths that lead to partly overlapping results. First, we have the international instruments, such as the 1948 Universal Declaration of Human Rights (UDHR), on which we shall focus in this contribution.¹ At the state level, human rights are translated into fundamental rights enshrined in national constitutions or in transnational instruments, such as the EU Charter of Fundamental Rights. The importance of human rights for the regulation of ITs is borne out by the increasing role that human rights discourse plays in the debate on Internet governance and on the regulation of AI, where such discourse provides common ground enabling all stakeholders (governments, economic actors, and civil society) to frame their different perspectives in the effort to find some convergence.²

2.3. The implementation of human rights in the IT domain

Developing and deploying ITs in ways that contribute to the implementation of human rights, rather than detracting from it, is a challenging task. On the one hand, tensions may exist between different valuable goals to be jointly pursued, i.e., between the need to satisfy conflicting rights (e.g., right to reputation and freedom of expression and information) or between rights and a social values (e.g., between privacy and national security, crime prevention, or scientific research), or between rights and market-based considerations (as with the need to protect investments or ensure consumer satisfaction or profitability). On

¹ Several other human rights instruments exist, among which the International Covenant on Civil and Political Rights and the International Covenant on Economic, Social and Cultural Rights. Among the regional treaties, we can mention the European Convention on Human Rights, the American Convention on Human Rights, and the African Charter on Human and Peoples' Rights.

² See the 2021 UN Human Rights Council [Resolution](#) on the promotion, protection, and enjoyment of human rights on the Internet (UN Doc. A/HRC/47/L.22).

the other hand, developing technological and social solutions that may solve such conflicts, or at least mitigate them, requires a combination of technological, social, and legal skills.

In legal theory, the idea that different, possibly conflicting rights and values are to be taken into consideration has been expressed by claiming that norms establishing fundamental rights are “optimisation” requirements rather than strict rules, and by distinguishing the broad scope of a right, covering whatever may contribute to its full enjoyment, from the limitations that may need to be placed on the protection of that right in concrete situations (see Alexy 2000; Barak 2012; Sartor 2018). On this approach, the protection of fundamental rights is to be governed by proportionality, according to which a fundamental right can be limited, but only to the extent necessary to realise a valuable interest, to an extent that outweighs the limitation of the right.

Until recently, proportionality-based assessments were only applied to measures adopted by governments. This was related to the idea that the law and the state’s apparatus should be in service of implementing fundamental rights and social values, translating their proportionate evaluation into precise legislative prescriptions addressed to private parties. However, proportionality-based assessments of fundamental-rights impacts have recently been extended to the design and deployment of any kind of IT solution, by both private and public entities. This development is linked to the acceleration of technological innovations and to their pervasive and often unexpected impacts in all domains of social life, which make it impracticable for legislation to ensure that fundamental rights are respected by establishing in advance what technologies are to be applied in what ways. EU law has indeed explicitly recognised that private actors need to directly take fundamental rights into account in developing and deploying ITs. We can see this in important instruments, such as the General Data Protection Regulation or the Artificial Intelligence Act: developers and deployers of technologies should ensure that interferences in fundamental rights are minimised and justified by overriding concerns.³

Similarly, in 2019 the Council of Europe Commissioner for Human Rights issued a recommendation titled “Unboxing Artificial Intelligence: 10 Steps to Protect Human Rights” under which AI applications deployed in the public domain need to be subjected to a human rights impact assessment. And in the United States in 2022 the White House put out a Blueprint for an AI Bill of Rights addressed to federal agencies.

2.4. Human rights, ITs, and equality

Since the 1980s the development of ITs has been accompanied by two parallel trends: the increase in economic inequality within countries and the growing importance of human rights.

³ Sartor 2023; Novelli et al. 2023; Mantelero 2022. For a guideline and questionnaire, see the Fundamental Rights and Algorithms Impact Assessment (FRAIA), issued in 2022 by the Dutch Ministry of the Interior.

Inequality in income and wealth has grown as a larger portion of the gross domestic product has gone to profits and rents rather than to wages and salaries, and as earning differentials between workers have increased. Multiple social, political, ideological, and economic factors play into this dynamic,⁴ but technological developments may also have contributed, in absence of countervailing social arrangements. Indeed, technological innovation creates a high demand for highly qualified jobs and increases the productivity of such jobs, while lowering demand for jobs that can be replaced by technology or devaluing jobs that get deskilled as a consequence of automation (see Brynjolfsson and McAfee 2014). At the same time, an increasing share of revenue from productive activities goes to capital rather than to labour, as technology substitutes labour or plays an increasing role in productivity. Moreover, technologies, by supporting scalability and reducing marginal costs, favour winner-take-all markets, where a single company or a few emerge as market leaders and extract monopolistic or oligopolistic prices.

We might wonder what role of the human rights discourse has played in this context of increasing economic inequality: has it countered this trend, however much with limited success, or, on the contrary, has it somehow favoured it, by replacing more effective appeals to social policies and equitable sharing.

The idea that human rights may contribute to inequality may seem paradoxical, given that human rights are equally attributed to all individuals, being grounded in their shared basic equality (equal dignity). However, what matters is not so much the affirmation of the abstract principle of equal concern and respect for every person but the question of “equality of what” (Sen 1997), i.e., the way in which the principle is translated into more specific normative requirements. The demand for equality under human rights may indeed support, or at least fail to counter, substantial inequality to the extent that the following obtains: (a) a minimalist human right approach is adopted that focuses on judicial protection against state encroachment on negative liberties or in any case reduces social rights to a minimum threshold to be universally guaranteed **or** (b) human rights are *equated* with social justice, through and through, and in this way the invocation of human rights replaces rather than complements broader calls for social justice. Appeals to human rights, also with regard to the deployment ITs, have indeed usually focused on the equal (or preferential) treatment of previously discriminated-against groups (such as women, LGBT people, and ethnic minorities), without challenging vast social differences in resources and life opportunities (Kocheno, Búrc, and Williams 2015; Moyn 2018).

A way to overcome this limitation may be to adopt the open and broad concept of human rights previously outlined (Section 2), on which human rights are understood as important, morally justified demands made in connection with individual freedoms, in turn understood as genuine opportunities. The extent to which these demands are to be realised may vary according to different moral and political views, which shape social choices, and thus legal implementations of human rights, according to the means available in different communities (this does not exclude the obligation that minimal levels be ensured for

⁴ See, among others, Stiglitz 2012 and Piketty 2020.

everybody). Human rights, thus broadly defined, are therefore a matter for a debate open to considerations of both inclusion and social justice, a debate that is not be restricted to the interpretation of existing legal sources. This debate should also include political and social strategies meant to ensure that everyone benefits from ITs, thanks to measures designed to afford greater opportunities, as by providing broader access to data, digital services and education, improving working conditions, and implementing wealth-redistribution policies.

3. ITs and human rights principles: freedom, dignity and equality

In this section, before turning to an analysis of individual human rights, we shall consider the fundamental principles of freedom, dignity, and equality as enshrined in Art. 1 UDHR: “All human beings are born free and equal in dignity and rights.” Similarly, the European Charter of Fundamental Rights recognises, in its preamble, the “indivisible, universal values of human dignity, freedom, equality and solidarity.”

3.1. Freedom

In freedom (understood as self-determination) and dignity (understood as the consideration and respect owed to each person) we have two fundamental, though largely undetermined, touchstones for assessing the opportunities and risks of ITs. Both of these principles are controversial and multifaceted, and different aspects may pull them toward opposite outcomes. Consider, for instance, how freedom of expression and dignity (in the form of the right to protection against attacks on reputation) may conflict in the context of online distribution of information.

We can say that freedom, comprehensively understood, includes one’s ability to achieve the outcomes one wants or has reason to want, and to do so through processes one endorses. ITs certainly have increased our sphere of freedom by expanding the range of valuable options available to us and the ways by which to implement these options. For instance, the Internet has expanded access to culture and education and has facilitated participation in economic and scientific activities. Similarly, AI offers additional ways to access content (e.g., through intelligent search and automated translation) and to process it, e.g., for purposes pertaining to entertainment, work, art, or science. The additional opportunities to access information and communicate also enhance the social dimension of autonomy, namely, the possibility to interact, achieve shared goals, and engage in political debate and action. The nondomination aspect of freedom—namely, the ability to exercise one’s freedom without depending on the power of others—is enhanced by the fact that basic IT technologies—to be used for activities such as communicating information and generating digital content, or even material objects, as through digital printing—are available at an affordable market price and without the need to obtain authorisations or permissions.

While the emphasis was until recently on the ways IT contributes to freedom—with benefits for large swaths of humanity, though to different extents—today the emphasis often falls on *interference* with freedom. ITs enable pervasive surveillance over individuals: all interactions with computer systems can in principle be recorded; physical behaviour

can be captured through all kinds of sensors, which are increasingly populating private and public environments; such personal data can be processed to infer further information (e.g., recognising faces, making classifications, ascribing sentiments and attitudes). Surveillance affects freedom, since it may induce people to refrain from activities that—once detected and linked to their identity—might expose them to adverse reactions. Freedom is further curtailed by profiling, i.e., through automated inference based on personal data, which may lead people to abstain from activities they think may lead to unfavourable inferences, possibly leading to adverse automated decisions (e.g., with regard to lending or employment). Automated assessments may also expose people to discourse-averse domination, as individuals may be subjected to automated judgments they cannot challenge through reasoned objections.

A further challenge to freedom pertains to hidden persuasion and manipulation, where people are nudged into options they would not have chosen without undue influence, or into forming opinions and sentiments that they would not otherwise have had. The possibilities for profiling and manipulation—and the chilling effect inherent in the mere possibility of adverse effects—increase to the extent to which personal or even group data are replicated and distributed as commodities within the complex ecology (involving data subjects data collectors, platforms, sellers, buyers, etc.) of so-called surveillance capitalism (Zuboff 2019). In authoritarian regimes, online censorship affects people’s ability to access, form, and express opinions. Some forms of IT-enabled censorship may also be present in liberal regimes, particularly through automated filtering.

Interferences with individual freedom may also be deliberate and personal. The enhanced communication and information capabilities arising from ITs can indeed be used to threaten, bully, or harass others, use aggressive or slanderous language, distribute unlawful or fake information, or otherwise create toxic digital environments, in which open and civil interaction no longer is possible. Under such conditions, those who are or may be negatively affected may abstain from engaging in the interactions at issue. It also appears that certain kinds of online aggression which may affect people’s ability to freely participate in online exchanges more often or damagingly affects certain groups, such as women, minorities or young people, even if no section of the population is exempted from such vicious attacks.

3.2. Dignity

Dignity is an even more controversial principle than liberty, as its understanding is based on multiple philosophical and political perspectives: it concerns what is valuable in each human being, and is thus worthy of respect (Barak 2013). It is invoked in order to support claims for expanding individual choice as well as for restricting such choice (e.g., in connection with enhancement technologies) (Beyleveld and Brownsword 2001).

Dignity is connected to some level of autonomy and would be negatively affected should “techno-regulation”—through surveillance, virtual constraints, and nudging—pre-empt self-determination. Dignity is also connected to the idea that humans should not be treated “merely” as means (according to Kant’s principle of humanity), which may happen where human behaviour is directed or influenced by computer systems for commercial, political,

or other purposes without any concern for the individual being targeted. From this perspective, manipulatory advertising which profits from people's vulnerability, or which directs them toward activities that may be seen as incompatible with respect for their dignity (e.g., where individuals are pushed into gambling or addiction). Dignity is also affected when people's discursive standing is not recognized (Pettit 2001), in particular where a person is the object of unchallengeable automated assessments. There also emerge issues of equality, as individuals in weaker socioeconomic conditions are more likely to be subject to unfavourable assessments and have a more limited capacity to react to them. An example can be found in a recent Dutch case where automated technologies with built-in assumptions were used to detect probable instances of child-benefit fraud, such probabilistic assessments automatically triggering requests for payment within strict deadlines. As a consequence a large number of parents were unjustly pressured into paying back the allowances they had received (which drove many poor families, often of immigrant background, into financial hardship).

An extreme violation of human dignity can also be found when humans are targeted by autonomous weapons, incapable of the ethical reasoning that is needed to respect humanity—incapable, that is, of implementing the principles of necessity, distinction, and proportionality established in humanitarian law.

3.3. Equality and nondiscrimination

Equality is affirmed not only by Art. 1 UDHR but also by Art. 7: "All are equal before the law and are entitled without discrimination to equal protection of the law." As noted, different aspects of equality are at stake in the deployment of technologies.

Many of the opportunities provided by ITs have increasingly become available to a larger number of people due to the expansion of communication networks, the diminishing price of devices, and an increased awareness of the significance of access to digital communications and services (consider, for instance, the explosion in Internet access and usage of mobile phones in developing countries). Thanks to ITs, a vast and increasing number of people, in many parts of the world, has had access to information, communication, and education. ITs have also facilitated engagement in economic activities (e.g., those linked to remote work) by making it possible to overcome geographical distances and impediments to mobility.

In recent year, there has been notable improvements in closing the so-called digital divide (the gap separating individuals and groups who can access and deploy ITs from those who cannot). For instance, most people in developed countries and an increasing proportion of those in developing ones now have access to the Internet. This is a huge achievement, obtained in a short time, but much remains to be done: digitally disadvantaged individuals should be provided with better connectivity, digital skills, and access to content and

services that are relevant to them.⁵ Some inequalities by social class, gender, age, disability, ethnicity, race, and location persist when it comes to the use of ITs, even if some gaps (such as that between male and female users) have significantly diminished.⁶

Equality is also at issue where the capabilities of ITs are put to use to collect, store, and process information for the purpose of treating people differently on the basis of classifications of them extracted from stored or inferred data. Here, too, we have a mixed picture. On the one hand, information technologies can be used to detect discrimination and build fairer systems, possibly combining automated and human assessments. On the other hand, IT systems can reproduce, and even expand, discriminatory practices, in commerce, healthcare, recruiting, the judicial system, etc. A huge discussion is taking place on how AI-based decision-making can contribute to or, on the contrary, detract from equality of opportunity (see Barocas, Hardt, and Narayanan 2023). With regard to machine-learning systems, it has been observed that different factors may lead to discriminatory outcomes: systems trained on prejudice-based or outdated examples may reproduce past injustices; individuals in less represented groups may be assessed less accurately, biased proxies for intended targets may favour certain classes or groups of people, and so on (Kleinberg et al. 2018). Different models may implement different visions of equality, from formal equality of opportunity (giving no weight to prohibited features or to proxies for them), to meritocratic ideals (based exclusively on predictions of the ability to perform the tasks at stake), to affirmative actions for disadvantaged individuals or groups. Thus, automated decision-making raises social equality issues involving moral, political, and legal considerations: they therefore reach beyond technical aspects and need to ultimately be addressed through an inclusive democratic debate.

Even where ITs are deployed in ways that appear to be justified by a permissible interest (such as efficiency and ultimately profit for private companies), the very fact that such technologies allow for much cheaper and pervasive automated sorting and individualised decision-making may put undue pressure on individuals, which may affect their psychological well-being and autonomy. As new and finer distinctions can be made—e.g., between more or less reliable borrowers, more or less needy buyers, more or less healthy insureds—people in disadvantaged social circumstances may be subject to additional harshness, even when the distinction is made on non-invidious grounds. Indeed, individuals in weaker socioeconomic circumstances are more likely to be subject to unfavourable assessments and have a more limited capacity to react to them.

The ability to probabilistically anticipate unfortunate circumstances (e.g., being at higher risk for a serious illness) may also lead to an adverse treatment of the individuals concerned in ways that further diminish the opportunities available to them. For instance,

⁵ Report of the UN Secretary-General, Progress made in the implementation of and follow-up to the outcomes of the World Summit on the Information Society at the regional and international levels. (UN Doc. A/78/62 E/2023/49).

⁶ Facts and Figures 2022: Latest on global connectivity amid economic downturn. ITU Hub. 30 Nov. 2022.

in the insurance business, the ability to predict future health conditions may harm those for whom such predictions are negative (they might be subject to higher costs and inferior coverage in insurance policies), while benefitting those for whom the prediction is positive. In this way, the egalitarian function of insurance—i.e., pooling resources to cope with risks to which everybody is in principle equally exposed—can be undermined.

4. ITs and specific human rights

In this section we shall consider some human rights that are most significantly impacted by IT applications.⁷

4.1. The Right to Privacy and Reputation

There are two related aspects to the right to privacy:⁸ (1) a person's ability to make autonomous decisions on the most intimate aspects of their life, such as sexuality, personal appearance, and attachments to others; (2) a person's ability to control the collection and processing of information about themselves. This second aspect, sometimes referred to as information privacy, comes to the fore in relation to the development of ITs and has indeed become the object of a distinct fundamental right, the right to the protection of personal data, in the EU Charter of Fundamental Rights (art. 8). According to the Charter, the protection of the latter right requires that the following conditions be satisfied: (1) data is processed fairly for specified purposes; (2) the processing is based on consent or other legal basis; (c) the individual concerned has the right to access the data to have it rectified; (d) an independent authority is entrusted with compliance. While these conditions reflect the regulation of data protection in the EU, they set a standard that has global significance, as many countries are converging toward the EU data protection model (though not yet the United States).

The right to information privacy is strongly and most directly affected by ITs, as they make it possible to capture and process a large and increasing amount of personal information, from which further personal information can be extracted, possibly on the basis of probabilistic predictions, on which decisions can be based. As noted above, the processing of personal data exposes the individuals concerned to risks stemming from the adverse use of such data, as well as the chilling effects resulting from the mere possibility of such risks arising. The processing of personal data may also have an impact on the right to reputation, as it leads to the construction of digital shadows of individuals on which basis both humans and machines can form opinions about the individuals, where higher-level representations or classifications are inferred (as when determining that one is or is not appointable or

⁷ For recent detailed accounts of the impact that AI has on human rights, see Quintavalla and Temperman 2023; Zavržnik and Simončič 2023; Nemitz. 2023.

⁸ Art. 12 UDHR: "No one shall be subjected to arbitrary interference with his privacy, family, home or correspondence, nor to attacks upon his honour and reputation."

creditworthy) and actions are taken accordingly (e.g., refusing a contract or a loan). Thus, the right to reputation becomes a right to (digital) identity, involving the claim to be represented in a computer system according to one's real and actual features or according to one's chosen image. This also supports a right to be forgotten, namely, the right that access to prejudicial or unwanted information, no longer reflecting the identity of the person concerned, be restricted (Sartor 2016).

The right to information privacy is also connected to equality and nondiscrimination, since limitations on the processing of personal data may prevent unjustified differentiations based on such data. Data-protection regulations also contribute to equality by restricting the processing of data that are most likely to lead to discrimination (such as information about ethnic origin or about someone's health), requiring that personal data be processed with a legal basis and for a legitimate purpose, providing data subjects with the ability to challenge any unlawful or unfair processing of their data.

Unequal socioeconomic circumstances and cultural backgrounds may also lead diminished levels of protection of the right to privacy (especially when this right is not supported by strong data protection): weaker individuals may be more susceptible to consent to any processing of their data in exchange for access to online services, or for small monetary or other rewards (e.g., discounts or participation in lotteries).

4.2. The rights to security of the person and to property

The right to security of the person, as set forth in the UDHR,⁹ protects physical integrity, which is affected by the deployment of IT-based weapons (Section 3.11), and also by IT failures leading to death or injury (as may happen in transportation, factories, or critical infrastructures). However, we may understand "security of person" as also covering the human mind, and not only one's "embodied" mind (what is inside a person's head) but also one's "extended mind," that is, the digital memories recording one's thoughts and memories, as well as the tools one uses to complement one's cognitive and communicative efforts. What could also be relevant in this connection, besides the idea of security, is the protection of property,¹⁰ covering all IT devices that are owned by individuals and associations of them, as well as the data stored in such devices (also in the cloud). A right to portability—to have one's data exported in a reusable format from the platforms where they are stored—may also be linked to a broader understanding of a person's security and property. A minimal requirement for equality in the protection of this right is that data security and portability need to be ensured for everybody, and not just for those who are able to pay for premium services.

⁹ Art. 3 UDHR: "Everyone has the right to life, liberty and security of person."

¹⁰ Art 17 UDHR: "Everyone has the right to own property alone as well as in association with others."

4.3. The rights to freedom of expression, information, and association

The freedoms of expression and information¹¹ have vastly profited from ITs. Thanks to such technologies, we can communicate all over the world and immediately access content from anywhere. In particular, the Internet has allowed for costless universal distribution of communication and information, through web pages, blogs, discussion groups, and other online ways of delivering information and intellectual creations. This expanded freedom to express oneself, and to inform and be informed, has often been countered by oppressive regimes, which have for purposes of political control have restricted the use of ITs, by blocking or filtering content, criminalising otherwise legitimate online expression, imposing liabilities on intermediaries, disconnecting unwanted users from Internet access, cyber-attacking unwanted sites, applying surveillance to IT infrastructures, and so on.

As noted above, a difficulty in ensuring the freedoms of expression and information stems from a dialectical tension that is intrinsic to these rights. The freedoms of expression and information in principle require that everybody be free to express and access whatever content in whatever way; however, certain views or ways of expressing them may undermine both other people's opportunity to freely express their opinion and the quality of the information that can be accessed. This is the case where aggressive or derogatory language is used; where offensive, demeaning, or discriminating views are expressed; and where false or unverified information is distributed. Under such conditions, individuals who disagree with leading opinions may be prevented from expressing their views (being afraid of adverse reactions), and these be impeded from exercising their freedoms. Others, especially those who lack adequate education, may be induced to endorse false premises, often leading to extreme views.

Technology, and in particular AI, can help in moderating online interactions—as by filtering out hate speech or downranking fake news—but the challenge of ensuring that this is done in accurate and nondiscriminatory ways are huge (automated filters may end up censoring legitimate speech, including minority opinions, or may prove to be inaccurate when dealing with less-spoken languages, for example) (De Gregorio and Dunn 2023). On the other hand, neither is it an acceptable approach, from a human rights and equality perspective, to remove all controls, thereby exposing individuals to unrestricted abuse and deceit in the name of so-called free-speech absolutism.

The opportunities for online interactions— through mailing lists, forums, social networks, and other online platforms— enhance freedom of association.¹² But this freedom, too, can unfortunately also be interfered with, by blocking or filtering content on websites

¹¹ Art. 19 UDHR: “Everyone has the right to freedom of opinion and expression; this right includes freedom to hold opinions without interference and to seek, receive and impart information and ideas through any media and regardless of frontiers.”

¹² Art. 20 UDHR: “Everyone has the right to freedom of peaceful assembly and association.”

maintained by associations, or by disrupting associative interactions, or again by subjecting their participants to unwarranted surveillance.

4.4. The right to an effective remedy

ITs can contribute to the right to an effective judicial remedy¹³ by supporting speedier and fairer judicial proceedings (as through digital access to legal information, workflow management, drafting tools, and decision-making support). However, there are worries about the predictive use of ICT in judicial procedures, specifically to forecast the outcome of legal cases or the behaviour of defendants. In the first regard, the fear is that a mechanical application of the law will be favoured, disregarding what justice requires in each individual case, while also neglecting the need to adapt the law to social change. In the second regard, it has been argued that automated prediction may lead to unfair or discriminatory outcomes, and in any case it may violate the requirement of individualised justice, since predictions about individuals are based on the statistical features of the groups to which they belong.

There are already predictive applications dealing with parole and recidivism, and their use may be detrimental to the individuals concerned, who may be subject to harsher treatment based on such predictions.¹⁴ The concern has been raised that such automated predictions may be discriminatory or may unduly tie people to their mistakes (being based on data referring to past behaviour). If the right to judicial protection is to be effectively protected, individuals adversely affected by an automated decision-making should be entitled to obtain a human review of decisions made about their case. In general, however, differential legal treatment based on predictions about people's future behaviour rather than on their past actions is questionable, especially under criminal law, since the practice may jeopardize the right to equality and nondiscrimination as well the presumption of innocence (in general, see Harcourt 2008).

The applications just considered may indeed negatively affect disadvantaged individuals, since their social circumstances (such as poverty, under-education, or unemployment) may lead to predictions which may be statistically justified (such circumstances being correlated with dangerous or criminal behaviour) but may nonetheless lead to additional hardship for such individuals (such as extended imprisonment or other curtailments of liberty). A differential treatment on this basis seems to be acceptable, on human rights grounds, only when consisting of or linked to educational and social measures (e.g., job

¹³ Art. 8 UDHR: "Everyone has the right to an effective remedy by the competent national tribunals for acts violating the fundamental rights granted him by the constitution or by law."

¹⁴ The COMPAS system, deployed in the United States, has been intensely debated. For a range of different perspectives on this question, see Larson et al. 2016; Dieterich, Mendoza, and Brennan 2016; Lagioia, Rovatti, and Sartor 2023.

training) meant to support the individuals concerned, so that they do not relapse into crime or other harmful behavior (Eijk 2020).

The right to a hearing, due process, and counsel are also at risk where a person is affected by automated decisions. Providing the individual concerned with appropriate explanations, and thus a sufficient degree of transparency, is a necessary precondition for the exercise of such rights.¹⁵ In implementing measures to satisfy these preconditions, we need to take into account the challenges that individuals face owing to socioeconomic circumstances or a cultural background that makes it more difficult for them to access those remedies.

4.5. Political rights

In the early days of the Internet there was a lot of optimism about the opportunities the new technology would create for democracy and political participation, including opportunities for citizens to take part in government.¹⁶ The Internet has indeed empowered individuals to freely engage with everybody, connecting with like-minded people and developing new bottom-up political initiatives. Moreover, communication between decision-makers and citizens has been supported and disintermediated thanks to the new digital channels.

A more sober perspective subsequently emerged. It has become apparent just how many difficulties get in the way of effective open online deliberation: time and attention constraints, intolerance toward different opinions, social pressure and aggressive behaviour, groupthink, among other factors. Moreover, some new challenges for the exercise of political rights have emerged.

For one thing, surveillance may undermine political rights. Pervasive data collection—by the state, but also by private actors that may collaborate with political authorities—may engender in individuals a fear of being subject to adverse actions, should unwanted behaviour be detected, or should circumstances be detected that provide new opportunities to harm them (as by attacking their reputation or anticipating their actions). Such adverse actions may include legal or social sanctions, the imposition of losses (dismissals, foreclosed opportunities), defamation, and unlawful and possibly violent attacks. Under such conditions, people may refrain from any activity that might possibly be detected and attract harmful reactions.

Secondly, a tendency toward polarisation has emerged, as people sharing certain views or social backgrounds choose to digitally interact only or preferably with people similar to them and to only access content they agree with (Sunstein 2020). This tendency is

¹⁵ This has been underscored by the UN Special Rapporteur on the Right to Privacy in the report titled “Principles of transparency and explainability in the processing of personal data in artificial intelligence” (UN Doc. A/78/310).

¹⁶ Art. 21 UCHR: “Everyone has the right to take part in the government of his country, directly or through freely chosen representatives.”

increased by the functioning of online platforms, which seek to attract users and keep them engaged by serving up information that confirms their views or otherwise reinforces their opinions (this being the content that people usually prefer and to which they can become “addicted”). The risks of falling prey to manipulation designed to influence the political sphere have increased, and they include the possibility of swaying voting through targeted messages, which may be AI-generated.¹⁷

Also, the very wealth amassed by leading IT companies raises new issues about the ability of citizens to effectively contribute to political process on an equal basis, as this wealth may also be used to influence the political process (through lobbying, and dark money, among other means) in ways that promote the interests of such companies. The ability to transfer assets and activities (and thus tax liabilities) to countries that offer more advantageous conditions also provides such companies with a formidable way of exercising pressure over political processes.

4.6. Social rights

ITs can have a big beneficial impact on social rights, for they can reduce the costs of managing and delivering social services, and so can contribute to the provision of a social safety net,¹⁸ as well as to an adequate standard of living.¹⁹ In areas such as healthcare, IT can be used, in combination with medical technologies, to provide new diagnostic and therapeutic means, thereby contributing to expanding the scope of the right to health. Efforts are needed not only to improve and expand the reach of these technologies, but also to develop cost-sustainable solutions and make them accessible to everybody. It is essential to take into account the risk that the algorithmic management of social security should lead to unfair or discriminatory outcomes or should otherwise negatively affect individuals or groups that are unable to make use of digital interfaces.

The UDHR does not explicitly include a right to live in a clean and healthy environment, which has been recognized in several national and international instruments.²⁰ We cannot here consider the complex relation between ITs and the environment. Let us just mention

¹⁷ The Cambridge Analytica case has provided an example of how this may happen with regard to the 2016 Trump election in the US and the 2016 Brexit referendum in the UK.

¹⁸ Art. 22 UDHR: “Everyone, as a member of society, has the right to social security and is entitled to realization [...] of the economic, social and cultural rights indispensable for his dignity and the free development of his personality.”

¹⁹ Art. 25 UDHR: “Everyone has the right to a standard of living adequate for the health and well-being of himself and of his family [...] and the right to security in the event of unemployment, sickness, disability, widowhood, old age or other lack of livelihood in circumstances beyond his control.”

²⁰ Examples are the UN resolutions adopted by the Human Rights Council in 2021 (A/HRC/RES/48/13) and the General Assembly in 2022 (A/RES/76/300).

that ITs may contribute to the preservation of the environment in multiple ways, as by monitoring the state of the environment, optimising energy and water consumption, providing new waste-management solutions, and supporting evidence-based environmental policies. On the other hand, the construction and deployment of ITs has sometimes come with considerable environmental harms. One is the pollution attendant on the extraction of the rare-earth minerals used in computer devices: these extractive industries directly affect the local communities, often without concern for the environmental degradation. Another is the energy consumption required for the IT infrastructure, and specifically for the operation and cooling of vast server farms or for training and deploying the largest AI models.

Finally, as noted above, ITs are relevant to the right to work.²¹ On the positive side, they create many new opportunities for gainful employment and may reduce occupational and workplace hazards by making it possible to monitor dangerous machines or relieve humans of the task of handling toxic or hazardous materials. On the negative side, ITs may restrict the opportunities of people engaged in activities that are taken over by automation, and who may consequently get demoted to menial and repetitive tasks. They may face unemployment or alienation, being deprived of the opportunity to engage in valuable and meaningful activities. In some contexts (e.g., logistics), it may also happen that human activities get confined to a few tasks that cannot (yet) be automated, where all workers' actions are controlled and directed by the automated system. The training of AI systems may also involve stressful and low-paid jobs (e.g., the labelling of data sets to classify and detect cases of hateful or violent content). As observed above, ITs can lead to greater disparity in the job market, widening the gap between highly skilled, highly paid workers and low-skilled, low-paid ones.

By facilitating the production and worldwide distribution of educational resources, ITs contribute as well to the right to education.²² Interactive learning tools can also reduce the cost of education and increase quality and accessibility, in the context of various e-learning or hybrid models. AI can play a significant role in improving the quality and flexibility of online courses. The risk here is that excessive reliance on automation and distance learning may, on the contrary, lower the quality of education by depriving students of the human relationship with teachers and other students.

Finally, ITs can facilitate the right to participate in culture and science:²³ they can do so by facilitating, to an unprecedented level, access to intellectual and artistic works, as well as the creation of new content. The risk here is that generative AI, in mining repositories of creative human content so as to generate substitutes for that content, may make it cheaper

²¹ Art. 23 UDHR: "Everyone has the right to work, to free choice of employment, to just and favourable conditions of work and to protection against unemployment."

²² Art. 26 UDHR: "Everyone has the right to education."

²³ Art. 27.1 UDHR: "Everyone has the right freely to participate in the cultural life of the community, to enjoy the arts and to share in scientific advancement and its benefits."

and easier to access content, but that output may be trite and standard, lacking in originality. The content so generated may harm authors, artists, and performers, and content creators generally, while at the same time flattening the cultural landscape.

This right is complemented by the rights protecting authors and inventors,²⁴ which brings us to the controversial issue of intellectual property rights. A vast literature exists on the extent to which IP rights, especially patents and copyrights, may limit access to useful products and content in ways that are especially detrimental to people who cannot afford to pay for such access, as well as to developing countries, holding back their progress. On the other hand, copyright still has a significant role to play in granting some protection to authors with regard to the digital exploitation of their work by IT companies — as where such works are made available on line or used to train generative AI system). Trade secrets are also relevant to human rights protection. They are so to the extent that these secrets may restrict access to information that is needed to check whether automated decision-making in any given case is correct and lawful, which in turn makes it possible to fight discrimination and unfair business practices.

5. Conclusion

I would like to conclude this analysis on a positive note. Which is to say that, as discussed, while ITs threaten human rights, they can also contribute to their protection and enhancement, providing them with new content. And human rights, in turn, may support the beneficial development and deployment of ITs and the equitable sharing of the advantages delivered by them. We should therefore view human rights not only as precious assets whose legacy needs to be protected against technological threats, but also as blueprints for a human-centred technological society. In conclusion, human rights provide partial yet valuable normative guidance for this purpose, across morality and law. They have indeed gained broad recognition, even if people differ over their content and relative weight. Clearly, they cannot offer an exhaustive plan for a good technological society, but are nonetheless crucial to its design. Indeed, if we are to establish such a society, it is essential that developers and deployers of ITs, at all levels, be aware of human rights and of their implications, so that they can work to actively build the very idea of human rights into the technological solutions they create and deploy.

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²⁴ Art. 27.2 UDHR: “Everyone has the right to the protection of the moral and material interests resulting from any scientific, literary or artistic production of which he is the author.”

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